

Gaussian Curvature and Some Applications



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Abstract

This research deal with Gauss curvature and some applications ,we study the curve on surface as a basic concept to discuss the importance and the role of curvature in natural.

In physics this becomes necessary in the applications of general relativity where it some times desired to calculate the Gauss curvature at a point in space from the observed geodesic phase of planets or light rays .and In engineering ,engineers involved in the design of structures such as geodesic domes frequently require a practical formula for computing the Gaussian curvature.

Introduction

Differential geometry is the study of geometry properties using differential and integral calculus.

In chapter one we study Gauss surface and geometry of curve ,parametrized curves, Regular curves arc length,The local theory of curve parametrized by arc length , serret frenet formulae , Global properties of plane curve(and his typs), Regular surface , The tangent plane the differential of a map, First and second fundamental form ,Orientation and Gauss map , The classification of point on S and some (Definitions ,Examples ,Theorems ,Properties)of Gauss surface .

Chapter two deals with curvature in Gauss map and Gauss curvature and Principle curvature , Mean curvature and for some examples and theorems.

Some applications given in chapter three of the gauss curvature in many propagation problems in physics and engineering.

Dedication

I dedicate this humble to the generous parents,Allah grant them health and wellness ,to everyone who contributed and supported this work, to all friends has not spared very much in helping me.

Chapter 1

Gauss surface

1.1 Geometry Of Curves

The Differential geometry of curves and surface has tow aspects one. which may be called classical differential geometry is the study of local properties curve and surface. By local properties we men those properties which depend only on the behaviour of the curve of surface in the neighborhood of a point the other aspect is the so.called global differential geometry here one studies the influence of the local properties on the behaviour of the entire curve or surface.

1.1.1 Parametrized curves

A parametrized differentiable curve is differentiable map $\alpha : I \longrightarrow R^3$ of an open interval $I = (a, b)$ of the real line R in to R^{3+} .

$\alpha(t) = (x(t), y(t), z(t)) \in R^3, t \in I$ the functions $x(t), y(t), z(t)$ are differentiable and t is the parameter of curve $\alpha(t)$ is trace.

if we denote by $x'(t)$ the first derivative of X at the point t and use similar notations for the functions Y and Z the vector $\alpha'(t) = (x'(t), y'(t), z'(t)) \in R^3$ is called tangent vector.

Example 1.1

The parametrized differentiable curve given by

$$\alpha(t) = (a \cos t, a \sin t, bt)$$

$\alpha'(t) = (-a \sin t, a \cos t, b)$ has differentiable and trace in R^3 it is called helix.

1.1.2 Regular curves Arc length

A parametrized differentiable curve $\alpha : I \rightarrow R^3$ is said to be regular if $\alpha'(t) \neq 0$ for all t belong I the arc length of a regular parametrized curve from the point t_0 is by definition $s(t) = \int_{t_0}^t |\alpha'(t)| dt$ where $|\alpha'(t)| = \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}$

in this case $\frac{ds}{dt} = 1 = |\alpha'(t)|$, that is velocity vectors has constant length equal to 1.

conversely, if $|\alpha'| \equiv 1$ then $S = \int_{t_0}^t dt = t - t_0$

The local theory of curves parametrized by arc length

let $\alpha : I = (a, b) \rightarrow R^3$ be a curve parametrized by arc length s . since the Tangent vector $\alpha'(s)$ has unit length, the norm $|\alpha''(s)|$ of the second derivative measures the rate of change of the angle which neighboring tangents make with the tangent at s . $|\alpha''(s)|$ gives, therefore, a measure of how rapidly the curve pulls away of s this suggests the following definition.

Definition 1.1

let $\alpha : I \longrightarrow R^3$ be a curve parametrized by arc length $S \in I$ the number $|\alpha'| = K(s)$ is called the curvature of α at s .

if α is a straight line , $\alpha(s) = us + v$,where u and v are constant vectors ($|u| = 1$), then $K \equiv 0$.

conversely ,if $K = |\alpha'(s)| \equiv 0$,then by integration

$\alpha(s) = us + v$, and the curves a straight line.

Serret frenet formulae

if $\alpha(t) = \alpha(s)$ be a regular curves parameter by arc length then we have

1. $t'(s) = Kn$
2. $n'(t) = -Kt - \tau b$
3. $b'(t) = \tau n$

1.1.3 Global properties of plane curves

- Simple curve : if there has no self intersection point .
- Closed curve : a curve in which the starting point and ending match is known as closed curve.
- Simple closed curve or four vertex : the curve has at least for vertex.

1.1.4 Regular surface

we shall now need some knowledge of calculus of several variables. specifically we need know some facts about continuity and differential of function and maps in R^2 and R^3 .

Definition 1.2

A subset $s \subset R^3$ is a regular surface if , for each $p \in s$ there exists a neighborhood v in R^3 and a map $x:u \longrightarrow v \cap s$ of an open set $u \subset R^2$ onto $v \cap s \subset R^3$ such that

1. x is differentiable ,this means that if we write $x(u,v)=(x(u,v),y(u,v),z(u,v)), (u,v) \in U$, the functions $x(u,v),y(u,v),z(u,v)$ have continuous partial derivatives of all order in U
2. x is a homeomorphism , since x is continuous this means that x has an inverse $x^{-1} : v \cap s \longrightarrow U$ which is continuous that is x^{-1} is the restriction of continuous $f : w \subset R^3 \longrightarrow R^2$ define on an open set W containing $v \cap s$
3. for each $q \in u$, the differential $dx : R^2 \longrightarrow R^3$ is one - to - one .

Example 1.2:

solution

let $x(u,v)$ and $(u,v) \in R^3$ $x(u,v) = (u,v,f(u,v))$ $X_u(u,v) = (1,0,f_u(u,v))$ $X_v(u,v) = (0,1,f_v(u,v))$ so $X(u,v)$ is differentiable and its image in R^3 is a simple surface.

Definition 1.3

An inner product on X is a mapping of $X \times X$ in to the scalar field K of X that is with $\langle X,Y \rangle$, and is called the inner product of X and Y , such that for all vector X,Y,Z and scalar α we have

1. $\langle X + Y, Z \rangle = \langle X, Z \rangle + \langle Y, Z \rangle$

$$2. \langle \alpha X, Y \rangle = \alpha \langle X, Y \rangle$$

$$3. \langle X, Y \rangle = \overline{\langle Y, X \rangle}$$

$$4. \langle X, X \rangle \geq 0 \quad \text{or} \quad \langle X, X \rangle = 0 \iff X = 0$$

1.1.5 The tangent plane , the differential of a map

In this section we shall show that , for each $q \in u$, the differential $dx_q : R^2 \longrightarrow R^3$ is one - to - one

for every $p \in s$ the set of tangent vector to the parametrized curves of s , passing through p , constitutes a plane

By a tangent vectors to s at a point $p \in s$,we mean the tangent vector $\alpha'(0)$ of a differentiable parametrized curve $\alpha : (-\epsilon, \epsilon) \longrightarrow s$ with $\alpha(0) = p$

Definition 1.4

By fixing a parametrization $X : U \subset R^2 \longrightarrow S$ at $p \in s$ we can make a definition choice of a unit normal vector at each point $q \in x(u)$ by the rule

$$N(q) = \frac{X_u \times X_v}{|X_u \times X_v|}$$

1.1.6 The First fundamental form

The first fundamental form tells how the surface inherits the natural inner product of R^3 .

Definition 1.5

let M be a regular surface in R^3 and let $p \in M$.the first fundamental form of M at p is the map

$$I_p = T_p M \times T_p M \longrightarrow R$$

$$(X, Y) \longrightarrow \langle x, y \rangle .$$

proof

$$\alpha(t) = \phi(u(t), v(t)), t \in (-\epsilon, \epsilon)$$

$$I_p(\alpha', \alpha') = \langle \phi_u * \frac{du}{dt} + \phi_v * \frac{dv}{dt}, \phi_u * \frac{du}{dt} + \phi_v * \frac{dv}{dt} \rangle$$

$$= \langle \phi_u, \phi_u \rangle \left(\frac{du}{dt}\right)^2 + 2 \langle \phi_u, \phi_v \rangle \frac{du}{dt} \frac{dv}{dt} + \langle \phi_v, \phi_v \rangle \left(\frac{dv}{dt}\right)^2$$

$$= E \left(\frac{du}{dt}\right)^2 + 2F \frac{du}{dt} \frac{dv}{dt} + G \left(\frac{dv}{dt}\right)^2$$

$$E = \langle \phi_u, \phi_u \rangle$$

$$F = \langle \phi_u, \phi_v \rangle$$

$$G = \langle \phi_v, \phi_v \rangle$$

$$g_{ij} = \begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix} = \begin{pmatrix} E & F \\ F & G \end{pmatrix}$$

1.1.7 Orientation and Gauss map

Orientation is a property of surface measuring whether is it possible to make a consistent choice for a normal vector at every point of the surface

Definition 1.6

A regular surface M is orientable if it possible to cover it with an atlas $\mathcal{A}$ so that

$$\forall i, j \in I, \forall p \in U_i \cap U_j : \det(\phi_j \circ \phi_i^{-1})(\phi_i(p)) > 0$$

A choice of an atlas that satisfies this condition is called an orientation of M , and M is called oriented.

a local chart (ϕ, U) is called a positive local chart, if no such atlas exists, M is called non-orientable.

Definition 1.7

let (M, m) be an oriented surface in R^3 and (ϕ, U) coordinate in U , the Gauss map in the map.

$$n : M \longrightarrow s^2 \subset R^3$$

$$P \longrightarrow n(p) = \frac{\phi_u x \phi_v}{|\phi_u x \phi_v|}(p)$$

the Gauss map is a differentiable map , it is differential induces the Weingarten - map
.

Definition 1.8

The Weingarten map is a linear map related to the Gauss map as follows the Weingarten map is the map $L_p : T_p M \longrightarrow T_p M$

$$V \longrightarrow -D_n(p)n = -\frac{d}{dt}(n \circ r)0$$

with $r(0) = p \in M$ and $v = r'(0) \in T_p M$.

Definition 1.9

The second fundamental form of a regular oriented surface M at a point $P \in M$ is the map

$$\prod_p : T_p M \times T_p M \longrightarrow R$$

$$(X, Y) \longrightarrow \langle L_p(x), Y \rangle$$

Proof

$$\begin{aligned} \prod_p(\alpha') &= - \langle D_n(p)(\alpha'), \alpha' \rangle \\ &= - \langle D_n(p)(\phi_u) \frac{du}{dt} + D_n(P)(\phi_v) \frac{dv}{dt}, \phi_u \frac{du}{dt} + \phi_v \frac{dv}{dt} \rangle \\ &= - \langle n_u \frac{du}{dt} + n_v \frac{dv}{dt}, \phi_u \frac{du}{dt} + \phi_v \frac{dv}{dt} \rangle \\ &= e \left(\frac{du}{dt} \right)^2 + 2f \frac{du}{dt} \frac{dv}{dt} + g \left(\frac{dv}{dt} \right)^2 \\ n_u &= D_n(P)(\phi_u) \end{aligned}$$

$$n_v = D_n(p)(\phi_v)$$

Problem 1.1

Given a parametrized of a hyperbolic parabolic $Z = X^2 - Y^2$ find the parametrized of a neighborhood U where $p \in u \subset s$

Solution

The parametrized of s is

$$X(u, v) = (u, v, u^2 - v^2)$$

$$X_u = (1, 0, 2u)$$

$$X_v = (0, 1, -2v)$$

$$X_{uu} = (0, 0, 2)$$

$$X_{vv} = (0, 0, -2)$$

$$X_{uv} = (0, 0, 0)$$

$$N = \frac{X_{ux}X_v}{|X_{ux}X_v|}$$

$$e = \langle X_{uu}, N \rangle = \frac{2}{(1-4u^2+4v^2)^{\frac{1}{2}}}$$

$$f = 0$$

$$g = \langle X_{vv}, N \rangle = \frac{-2}{(1-4u^2+4v^2)^{\frac{1}{2}}}$$

$$\prod_p(w) = eu^2 + 2fv'v' - gv' = 0$$

$$\frac{2u'2}{(1-4u^2+4v^2)^{\frac{1}{2}}} - \frac{2v'2}{(1-4u^2+4v^2)^{\frac{1}{2}}} = 0$$

$$2u'2 - v'2 = 0, u'2 - v'2 = 0$$

$$(u' - v')(u' + v') = 0, u' - v' = 0$$

$$u' + v' = 0$$

$$u - v = 0, u + v = 0$$

Theorem 1.1

Let $S \subset R^3$ be a regular surface and $P \in S$ there is a neighborhood V of P in S such that V the graph of a differentiable function which one of the forms

$$z = f(x, y), y = g(x, y), x = h(x, y)$$

$$g : U \subset \mathbb{R}^2 \longrightarrow S \subset \mathbb{R}^3$$

$$(\pi \circ g) : U \subset \mathbb{R}^2 \longrightarrow S \subset \mathbb{R}^3 \longrightarrow (x, y)$$

Proof

let $g : U \subset \mathbb{R}^2 \longrightarrow S \subset \mathbb{R}^3$ be a parametrized of s in p , $g(u, v) = (x(u, v), y(u, v), z(u, v))$

by condition (g is differentiable) and $dg_q(0)$ one of the determinate $\frac{\partial(x, y)}{\partial(u, v)}, \frac{\partial(y, z)}{\partial(u, v)}, \frac{\partial(x, z)}{\partial(u, v)}$

not equal zero at $g^{-1}(p) = q$ suppose $\frac{\partial(x, y)}{\partial(u, v)} \neq 0$ and consider the projection map

$$\pi : v_1 \subset s \longrightarrow (x, y)$$

the $\pi \circ g : U \subset \mathbb{R}^2 \longrightarrow (x, y)$ we apply the inverse function theorem to

obtain a neighborhood v_2 differentiable to v_1 it following that $(\pi \circ g)^{-1} : v_2 \longrightarrow v$

is 1-1 and homomorphism there for we teak $(x, y) \mapsto (u, v)$ and the

function $z = z(u, v) | x(u, v), y(u, v)$ is a differentiable surface .

Example 1.3

Show that the cylinder $(x, y, z) \in \mathbb{R}^3; X^2 + Y^2 = 1$ is a regular surface .

solution

let $X_1 : U \subset \mathbb{R}^2 \longrightarrow \mathbb{R}^3$ given by $X_1(x, y) = (x, +\sqrt{1-y^2})$, $(x, y) \in U$

where $\mathbb{R}^2 = [(x, y, z) \in \mathbb{R}^3; Z = 0]$ and $U = [(x, y) \in \mathbb{R}^2; X^2 + Y^2 < 1]$ is

a parametrization of s^2 , observe that $X_1(U)$ is the (open) part of s^2 above the xy plane.

since $x^2 + y^2 < 1$, the function $+\sqrt{1-y^2}$ has continuous partial derivatives

of all orders . thus X_1 is differentiable and condition 1 holds .condition 3 it

easily verified , since $\frac{\partial(x, y)}{\partial(x, y)} \equiv 1$ to check condition 2 , we observe that X_1

is on-to-on and that X_1^{-1} is the restriction of the (continuous) projection.

$\pi(x, y, z) = (x, y)$ to the set $X_1(U)$. thus X_1^{-1} is continuous in $X_1(U)$ we

shall now cover the whole cylinder with similar parametrizations as follows ,we define

$X_2 : U \subset \mathbb{R}^2 \longrightarrow \mathbb{R}^3$ by $X_2(x, y) = (x, -\sqrt{1-y^2})$ check X_2 is a parametrization and ob-

serve that $X_1(U) \cup X_2(U)$ covers S^2 minus the equator

$[(x, y, z) \in \mathbb{R}^3; X^2 + Y^2 = 1, Z = 0]$ then , using the x and y plane , we define the parametrizations

$$X_3(y, x) = (y, +\sqrt{1-x^2})$$

$$, X_4(y, x) = (y, -\sqrt{1-x^2})$$

then the cylinder $[(x, y, z) \text{ in } R^3; x^2 + y^2 = 1]$ is a regular surface.

1.1.8 The classification of point on a surface S

A point of a surface s is called

1. Elliptic if $\det(dN_p) > 0$
2. Hyperbolic if $\det(dN_p) < 0$
3. parabolic if $\det(dN_p) = 0$
4. planar if $\det dN_p = 0, \text{ with } dN_p \neq 0$
5. Umbilical points if $K_1 = K_2, \text{ the planar } (K_1 = K_2 = 0)$ are umbilical.

Theorem 1.2

If all point of sphere or in a plane .

proof

Let $p \in s$ and let $x(u, v)$ be a parametrization of s at p such that the coordinate neighborhood v is connected .

since each $q \in v$ is an umbilical point , we have for any vector $w = a_1x_u + a_2x_v$ in $T_q(s)$, $dN(w) = \lambda q(w)$ where $\lambda = \lambda(q)$ is a real differentiable . function in v . we first show that $\lambda(q)$ is constant in v ,for that we write the above equations as $N_ua_1 + N_va_2 = \lambda(X_ua_1 + X_va_2)$

hence ,since w is arbitrary $N_u = \lambda X_u, N_v = \lambda X_v$ differentiating the first equations in U and the second equation in V and subtracting the resulting equations , we obtain $\lambda_u X_v - \lambda_v X_u = 0$ since X_u and X_v are linearly independent

,we conclude that $\lambda_u = \lambda_v = 0$ for all $q \in v$. since v is connected , λ is constant in v , as we claimed if $\lambda \equiv 0, N_u = N_v = 0$ and therefore $N = N_0 = \text{constant}$ in v . thus $\langle x(u, v), N_0 \rangle = \langle x(u, v), N_0 \rangle = 0$ hence $\langle x(u, v), N_0 \rangle = \text{constant}$ and all point $x(u, v)$ of v belong to a plane .

if $\lambda \neq 0$, then the point $x(u, v) - (\frac{1}{\lambda})N(u, v) = y(u, v)$ is fixed , because

$$(x(u, v) - \frac{1}{\lambda}N(u, v))_u = (x(u, v) - \frac{1}{\lambda}N(u, v))_v = 0$$

since $|X(u, v) - y|^2 = \frac{1}{\lambda^2}$.all points of v are contained in a sphere of center y and radius $\frac{1}{|\lambda|}$.

Example 1.4

1. At an elliptic point the Gaussian curvature is positive . both principal curvature have the same sign .the point of a sphere are elliptic points .the point $(0,0,0)$ of the parabolic $z = x^2 + ky^2, k > 0$ is also an elliptic point
2. At a hyperbolic point ,the Gaussian curvature is negative . the principal curvature have opposite signs .the point $(0,0,0)$ of the hyperbolic parabolic $z = y^2 - x^2$ is a hyperbolic point.
3. At a parabolic point ,the Gaussian curvature is zero , but one of the principal curvature is not zero , the points of a cylinder are parabolic points.
4. At a planar point, all principal curvature are zero ,the points of a plane trivially satisfy this condition , Anon trivial example of a planar point was given $z = y^4$ about the z axis.

Chapter 2

The curvature in Gauss map

2.1 Gauss curvature

The fundamental idea behind the Gaussian curvature is the Gauss map , the Gaussian curvature can be define a follows .

Definition 2.1

The Gaussian curvature of the regular surface M at a point $p \in M$ is $k(p) = \det(D_n(p))$. where $D_n(p)$ is the differential of the Gauss map at p , it holds that for a local parametrization $\phi(u, v)$ and a curve $\alpha(t) = \phi(u(t), v(t))$ with $\alpha(0) = p \in M$ the tangent vector to the curve α at $t = 0$ equals

$\alpha'(0) = \phi_u(u(0), v(0)) \frac{du}{dt}(0) + \phi_v(u(0), v(0)) \frac{dv}{dt}(0)$ and thus $D_n(\alpha'(0)) = n_u \frac{du}{dt}(0) + n_v \frac{dv}{dt}(0)$ since n_u and n_v lie in the tangent plane $T_p M$, we can write them in terms of the basis (ϕ_u, ϕ_v) as $n_u = \alpha_1 \phi_u + \alpha_2 \phi_v$ and $n_v = \alpha_3 \phi_u + \alpha_4 \phi_v$ for some $(\alpha_{ij})_{i,j=1,2} \in \mathbb{R}$ which

gives the matrix of the linear map $D_n(p)$

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{pmatrix}$$

we can now link the coefficients of the first fundamental form to those of the second fundamental form

since $\langle n, \phi_u \rangle = \langle n, \phi_v \rangle = 0$ it holds that

$$\begin{aligned} -e &= \langle n_u, \phi_u \rangle = \langle \alpha_{11}\phi_u + \alpha_{21}\phi_v, \phi_u \rangle = \alpha_{11}E + \alpha_{21}F - f = \langle n_u, \phi_u \rangle = \langle \alpha_{11}\phi_u + \\ \alpha_{21}\phi_v, \phi_v \rangle &= \alpha_{11}F + \alpha_{21}G - f = \langle n_v, \phi_u \rangle = \langle \alpha_{12}\phi_u + \alpha_{22}\phi_v, \phi_u \rangle = \alpha_{12}E + \alpha_{22}F \\ -g &= \langle n_v, \phi_v \rangle = \langle \alpha_{12}\phi_u + \alpha_{22}\phi_v, \phi_v \rangle = \alpha_{12}F + \alpha_{22}G \end{aligned}$$

which written in matrix form gives

$$-\begin{pmatrix} e & f \\ f & g \end{pmatrix} = \begin{pmatrix} \alpha_{11} & \alpha_{21} \\ \alpha_{21} & \alpha_{22} \end{pmatrix} \begin{pmatrix} E & F \\ F & G \end{pmatrix}$$

it holds that $EG - F^2 > 0$ since the inner product is positive definite and that $\det(A.B) = \det(A).\det(B)$ thus we get $K = \det(\alpha_{ij})_{i,j} = 1, 2 = \frac{eg-f^2}{EG-F^2}$ we will now study how the normal curvature at a point varies when a unit tangent vector varies .

in general , we will see that the normal curvature has a maximum value K_1 and a minimum value K_2 ,and that the corresponding direction are orthogonal .

the quantity $K = K_1K_2$ call the Gaussian curvature .we will compute K in terms of the first and the second fundamental form .

Recall that a surface X and some point P on X the vector X_u, X_v form a basis of the tangent space $T_p(x)$ given a unit vector $t^\rightarrow = X_{ux} + X_{vy}$,the normal curvature is given by $KN(t^\rightarrow) = Ex^2 + 2Fxy + Gy^2$ since $Ex^2 + 2Fxy + Gy^2 = 1$ usually (X_u, X_v) is not an orthonormal frame ,and it is useful to replace the frame X_u, X_v with an orthonormal frame one verifies easily that the frame (e_1, e_2) defined such that $e_1^\rightarrow = \frac{X_u}{\sqrt{E}}, e_2^\rightarrow = \frac{EX_v - FX_u}{\sqrt{E(EG-F^2)}}$ is indeed an orthonormal. with respect to this frame every unit vector can be written as $t^\rightarrow = \cos\theta e_1 + \sin\theta e_2$ and expressing (e_1, e_2) in terms of X_u and X_v we have $t^\rightarrow = (\frac{w\cos\theta - F\sin\theta}{w\sqrt{E}})X_u + (\frac{\sqrt{E}\sin\theta}{w})X_v$ where $w = \sqrt{EG - F^2}$ we can now compute $KN(t)$, and we get

$$KN(t^\rightarrow) = L(\frac{w\cos\theta - F\sin\theta}{w\sqrt{E}})^2 + 2M(\frac{(w\cos\theta - F\sin\theta)\sin\theta}{w^2}) + N(\frac{E\sin^2\theta}{w^2})$$

we leave as an exercise to show that the above expression can be written as $KN(t^\rightarrow) = H + A\cos 2\theta B\sin 2\theta$

$$\text{where } A = \frac{(L(EG-2F^2)+2EFM-E^2N)}{2E(EG-F^2)}, B = \frac{EM-FL}{E\sqrt{EG-F^2}}, C = \sqrt{A^2 + B^2}$$

Theorem 2.1

A necessary and surface condition for a connected regular curve on S to be a line of curvature of S is that $N'(t) = \lambda(t)\alpha'(t)$ where $\lambda(t)$ is a differentiable function

proof

1. let $c \in s$ be a line of curvature we prove that $N'(t) = \lambda(t)\alpha'(t)$ since c is line of curvature the $\alpha'(t)$ is a principal direction $dN(\alpha'(t)) = K\alpha'(t)$, $K(t) = \lambda(t)$ then c is line of curvature
2. let $N'(t) = \lambda(t)\alpha'(t)$ we prove that c is a line of curvature $dN(\alpha'(t)) = N'(t) = \lambda(t)\alpha'(t)$ hence $\alpha'(t)$ is a principal of direction

Example 2.1

we shall compute the Gaussian curvature of the points of the toros covered by the parametrization $X(u, v) = ((a + r \cos u) \cos v, (a + r \cos u) \sin v, r \sin u)$

$$0 < u < 2\pi \quad , \quad 0 < v < 2\pi$$

Solution

for the computation of the coefficients e, f, g we need to know N and thus X_u and $X_v, X_{uu}, X_{vv}, X_{uv}$

$$X_u = (-r \sin u \cos v, -r \sin u \sin v, r \cos u)$$

$$X_v = (-(a + r \cos u) \sin v, (a + r \cos u) \cos v, 0)$$

$$X_{uu} = (-r \cos u \cos v, -r \cos u \sin v, -r \sin u)$$

$$X_{uv} = (r \sin u \sin v, -r \sin u \cos v, 0)$$

$$X_{vv} = (-(a + r \cos u) \cos v, -(a + r \cos u) \sin v, 0) \text{ from these , we obtain .}$$

$$E = \langle X_u, X_u \rangle = r^2$$

$$F = \langle X_u, X_v \rangle = 0$$

$$G = \langle X_v, X_v \rangle = (a + r \cos u)^2 \text{ introducing the values just obtained in}$$

$$e = \langle N, X_{uu} \rangle \text{ we have since } |X_u \times X_v| = \sqrt{EG - F^2}$$

$$e = \langle \frac{X_u \times X_v}{|X_u \times X_v|}, X_{uu} \rangle = \frac{\langle X_u, X_v, X_{uu} \rangle}{\sqrt{EG - F^2}} = \frac{r^2(a + r \cos u)}{r(a + r \cos u)} = r$$

similarly we obtain

$$f = \frac{(X_u, X_v, X_{uv})}{r(a+r \cos u)} = 0$$

$$g = \frac{(X_u, X_v, X_{vv})}{r(a+r \cos u)} = \cos u(a + r \cos u)$$

$$\text{finally since } K = \frac{(eg-f^2)}{EG-F^2} \text{ we have that}$$

$$K = \frac{\cos u}{r(a+r \cos u)}$$

2.1.1 The Principal Curvature

Definition 2.2

Given a surface X , for any point p on X , letting A, B, H be define as above, and $C = \sqrt{A^2 + B^2}$ unless $A = B = 0$, the normal curvature K_n at p takes a maximum value K_2 called principal curvatures at p , where $K_1 = H + C$ and $K_2 = H - C$ the directions of the corresponding unit vector are called the principal direction at p . hence $K_1 = \frac{e}{E}$ and $k_2 = \frac{g}{G}$

Theorem 2.2

let M be an oriented regular surface in R^3 with Gauss map $N : M \rightarrow S^2$ and $p \in M$ then $z \in T_p M$ is a principal direction at p if and only if it is an eigenvector for the shape operator $S_p : T_p M \rightarrow T_p M$. let $[z_1, z_2]$ be an orthonormal basis for the tangent plane $T_p M$ of eigenvector to S_p $S_p(z_1) = \lambda_1 z_1$ and $S_p(z_2) = \lambda_2 z_2$, for some $\lambda_1, \lambda_2 \in R$ then every unit vector $z \in T_p M$ can be written as $z(\theta) = \cos \theta \cdot z_1 + \sin \theta \cdot z_2$ and $K_n(z(\theta)) = \langle S_p(\cos \theta z_1 + \sin \theta z_2), \cos \theta z_1 + \sin \theta z_2 \rangle = \cos^2 \theta \langle S_p(z_1), z_1 \rangle + \sin^2 \theta \langle S_p(z_2), z_2 \rangle + \cos \theta \sin \theta (\langle S_p(z_1), z_2 \rangle + \langle S_p(z_2), z_1 \rangle) = \lambda_1 \cos^2 \theta + \lambda_2 \sin^2 \theta$ if $\lambda_1 = \lambda_2$ then $K_n(z(\theta)) = \lambda_1$ for all $\theta \in R$ so any direction is both principal an eigenvector for the shape operator S_p .

if $\lambda_1 > \lambda_2$ then we can assume, without loss of generality, that $\lambda_1 \neq \lambda_2$.

then $z(\theta)$ is a maximal principal direction if and only if $\cos^2 \theta = 1, z = \pm z_1$ and clearly a minimal principal direction if and only if $\sin^2 \theta = 1, z = \pm z_2$

Example 2.2

consider the parametrized surface $X(u, v) = (u - \frac{u^3}{3} + uv^2, v - \frac{v^3}{3} + vu^2, u^2 - v^2)$ compute the principal curvature .

solution

$$X_u = (1 - u^2 + v^2, 2uv, 2u)$$

$$X_v = (2uv, 1 - v^2 + u^2, -2v)$$

$$X_{uu} = (-2u, 2v, 2)$$

$$X_{vv} = (2v, 2u, -2)$$

$$X_{uv} = (2v, 2u, 0)$$

$$E = \langle X_u, X_u \rangle = (1 + u^2 + v^2)^2$$

$$G = \langle X_v, X_v \rangle = (1 + u^2 + v^2)^2$$

$$F = \langle X_u, X_v \rangle = 0 \quad N = \frac{X_u \times X_v}{|X_u \times X_v|} = (0, 0, 1)$$

$$e = \langle X_{uu}, N \rangle = 2$$

$$g = \langle X_{vv}, N \rangle = -2$$

$$f = \langle X_{uv}, N \rangle = 0$$

$$K_1 = \frac{e}{E}, K_2 = \frac{g}{G}$$

$$K_1 = \frac{2}{(1+u^2+v^2)^2}$$

$$K_2 = \frac{-2}{(1+u^2+v^2)^2}$$

2.1.2 The Mean Curvature**Definition 2.3**

the average $H = \frac{K_1 + K_2}{2}$ of the principal curvature is called the mean curvature .

observe that the principal direction θ_0 and $\theta_0 + \frac{\pi}{2}$ corresponding to K_1 and K_2 are orthogonal . after some laborious calculations we get the following (famous) formulae

$$\text{for the mean curvature } H = \frac{GL - 2FM + EN}{2(EG - F^2)}$$

Example 2.3

Let $z = A(u, v)$ belong to an open set $U \subset \mathbb{R}^2$ it is therefore , convenient to have at hand formulas for the relevant concepts in this case . obtain such formula let us parametrized the surface by

$$X(u, v) = (u, v, A(u, v)), \quad (u, v) \in U$$

where $u=x$ and $v=y$. a simple computation show that

Solution

$$X_u = (1, 0, A_u)$$

$$X_v = (0, 1, A_v)$$

$$X_{uu} = (0, 0, A_{uu})$$

$$X_{uv} = (0, 0, A_{uv})$$

$$X_{vv} = (0, 0, A_{vv})$$

thus

$$N(x, y) = \frac{(-A_x, -A_y, 1)}{(1+A_x^2+A_y^2)^{\frac{1}{2}}}$$

$$E = \langle X_u, X_u \rangle = (1, 0, A_x^2)$$

$$G = \langle X_v, X_v \rangle = (0, 1, A_y^2)$$

$$F = \langle X_u, X_v \rangle = (0, 0, A_x A_y)$$

$$e = \frac{A_{xx}}{(1+A_x^2+A_y^2)^{\frac{1}{2}}}$$

$$f = \frac{A_{xy}}{(1+A_x^2+A_y^2)^{\frac{1}{2}}}$$

$$g = \frac{A_{yy}}{(1+A_x^2+A_y^2)^{\frac{1}{2}}}$$

$$H = \frac{(1+A_x^2)A_{yy} - 2A_x A_y A_{xy} + (1+A_y^2)A_{xx}}{(1+A_x^2+A_y^2)^{\frac{3}{2}}}$$

Chapter 3

Some Applications in Gaussian curvature

Gauss surfaces work can be generalized to n -dimensional manifolds, this leads to use curvature of space time in general relativity .

Gauss surface is very important for the development in modern physics.

Carl Friederich Gauss (1777-1855). Gauss's original work is currently in print by Dover Publications (Mineola, NY 2005) in *General Investigations of Curved Surfaces*. This brief book (130 pages) includes the 1827 *General Investigations of Curved surfaces* (*Disquisitiones generales circa superficies curvas*) and the 1825 *New General Investigations of Curved Surfaces*. There are several pages of notes and additional notes. The Dover book is a reprint of the Princeton University Library version of 1902 which was translated from the original Latin and German by Adam Hiltebeitel and James Morehead. A historical introduction (on which the historical material in these notes is based) is given by Peter Pesic. The images for this section are from the MacTutor History of Mathematics archive. Gauss's work on surfaces was generalized by his student, Georg Friederich Bernhard Riemann (1826-1866), to n -dimensional manifolds.

This laid the foundations needed for Albert Einstein (1879-1955) to use curvature of spacetime in his work on general relativity.

According to Einstein, the importance of Gauss for the development of modern physical theory and especially for the mathematical fundamentals of the theory of relativity is overwhelming indeed; . . . If he had not created his geometry of surfaces, which served Riemann as a basis, it is scarcely conceivable that anyone else would have discovered it.

3.1 Applications to physics

Light is an electromagnetic wave and thus exhibits polarization its waves can oscillate in more than one direction. When unpolarized light, that is, light with no defined polarization, passes through a polarizer, only waves of a certain polarization are retransmitted (the waves that are in the plane of polarization) and waves of different polarizations are blocked. Two polarizers used together transmit light differently depending on their relative orientation. For example, when the polarizers are arranged so that their planes of polarization are perpendicular to each other, the light is blocked. When the second filter (called the analyzer) is parallel to the first, all of the light passed by the first filter is also transmitted by the second. This means that we can have a singularity with index $\frac{1}{2}$. Experimental model systems of spherical nematics have shown that the thickness of the shell plays a crucial role in the distribution and index of the singularities of a field of nematics. As an example, for a field of nematics with two singularities, there are two possible configurations depending on the homogeneity of the thickness of the shell. If the shell is rather thick and homogeneous over the whole sphere, then the singularities tend to align in such a way that the distance between them is maximized, that is, diametrically. But, when increasing the thickness inhomogeneity towards a critical value, one defect migrates towards the

other one such that the singularities are all confined on the thinnest part of the shell.

The link to the mathematics of surfaces comes from the theorem of Poincaré-Hopf which tells us that the sum of the indices, that is, the singularities of a vector field on a sphere is always equal to two. This means that the following configurations can theoretically occur:

- 1 x index 2
- 2 x index 1
- 1 x index 1 + 2 x index $\frac{1}{2}$
- 1 x index $\frac{3}{2}$ + 1 x index $\frac{1}{2}$
- 4 index $\frac{1}{2}$

The cases involving an index of 2 or $\frac{3}{2}$ are not likely to occur, due to energetical reasons. In the experiment above, unpolarized light is sent through a polarizer and an analyzer positioned with their axes of polarization perpendicular to each other. The nematic droplet is used as an intermediate medium between the two polarizers. The nematic director of the nematic is uniform and makes an angle θ with the axis of the first polarizer. If the angle θ is 0, the nematic does not change the polarization of the light and thus the light is blocked. If the angle θ is $\frac{\pi}{2}$ the polarized light is blocked by the nematic and no light passes. Finally, if θ is some other angle different from 0, $\frac{\pi}{2} \pmod{\pi}$, the nematic changes the polarization of the light and light passes through the second polarizer and can be detected. Using this last setup, simply counting the number of brushes (enlightened parts) at each defect and dividing it by 4 gives its index. (We get two brushes at a defect of index $\frac{1}{2}$ and four brushes at a defect of index 1).

3.1.1 Calculation of the Gaussian Curvature From Geodesic Deviation

It is well known that any general 2-dimensional surface is topologically equivalent at any given point to a surface of revolution. Hence, two close geodesics on the surface, when considered only within a small surface patch, can be treated as embedded in a surface of revolution. Such curves, however, will not necessarily be geodesics in the surface of revolution. Consider now a surface of revolution, where the smooth curve S is a general curve that is embedded in the surface. S' is a copy of S that is obtained by rotating S through a small angle θ . p^r is a position vector, defined over a circular ring passing by S - S' . Let us select two parameters u and v , such that u varies as we travel along the curve S , but v is constant, and v varies as we pass from one curve to another, but u is constant. Obviously

$u = s, v = \theta$ where θ is the rotation angle of the axis from S to S' . Given such parameters on any surface in space,

$$G = \frac{(1/2)d^2p^2/ds^2 - 2(1 - \cos \psi)}{[(dp^2/ds)/2(1 - \cos \psi)]^2 - p^2}$$

where G is the Gaussian curvature of the surface at the location of the vector P .

3.1.2 Examples of Applications in Engineering

Recall if $f(x, y, z)$ is (scalar valued) function, then for C a constant $f(x, y, z) = c$ determines a surface (we assume all second partial of f are continuous and so the surface is smooth) the gradient of f is

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

if $v^{\rightarrow}$ is a vector tangent to the surface $f(x, y, z) = c$ at point $p_0^{\rightarrow} = (x_0, y_0, z_0)$ then

$\nabla f(x_0, y_0, z_0)$ is orthogonal to $v_0 \vec{v}$ (and so ∇f as the normal vector for the plane).

Example 3.1

Find the section of $M : x^2 + y^2 = 1$ (an infinity tall right circular cylinder of radius 1) at the point $\vec{p} = (1, 0, 0)$ in the direction $\vec{v} = (0, 1, 0)$.

Solution

A normal vector to M at $\vec{p}$ is

$$\nabla(x^2 + y^2) = (2x, 2y, 0)|_{(1,0,0)} = (2, 0, 0)$$

therefore we take $\vec{U} = (1, 0, 0)$ the plane containing $\vec{U}$ and $\vec{v}$ has a normal vector

$$\vec{U} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = (0, 0, 1)$$

therefore the equation of this plane is

$$0(x - 1) + 0(y - 0) + 1(z - 0) = 0$$

or $z=0$, the intersection of this plane and the surface is

$$\vec{\alpha}_{\vec{v}} = [(x, y, z) | x^2 + y^2 = 1, z = 0]$$

Definition 3.1

Let $\vec{\alpha}_{\vec{v}}$ a normal section to a smooth M at point $\vec{p}$ in the direction $\vec{v}$ let $\vec{V}$ be a unit normal to M at $\vec{p}$ ($\vec{U}$ is also a unit normal to M at $\vec{p}$) the normal curvature of M at $\vec{p}$ in the $\vec{v}$ direction with respect to $\vec{U}$, denoted $K_{n,\vec{U}}(\vec{v})$ is

$$K_{n,\vec{U}}(\vec{v}) = \frac{\vec{U} \cdot \vec{N}}{R(\vec{v})}$$

where $\vec{N}$ is the principal normal vector of $\vec{\alpha}_{\vec{v}}$ at $\vec{p}$. if $\vec{\alpha}_{\vec{v}}$ has zero curvature at $\vec{p}$, we take $k_{n,\vec{U}}(\vec{v}) = 0$.

Example 3.2

what is $K_{n,\vec{U}}(\vec{v})$ for the cylinder $x^2 + y^2 = 1$ at $\vec{p} = (1, 0, 0)$ in the direction $\vec{v} = (0, 1, 0)$ with respect to $\vec{U} = (1, 0, 0)$

Solution

As we saw in the previous example ,

$$\vec{\alpha}_v = [(x, y, z) | x^2 + y^2 = 1, z = 0]$$

we can parametrized $\vec{\alpha}_v$ as

$$\vec{\alpha}(s) = (\cos s, \sin s, 0)$$

where $s \in [0, 2\pi]$, then

$$\vec{T}(s) = \vec{\alpha}'(s) = (-\sin s, \cos s, 0)$$

and

$$\vec{T}'(s) = \vec{\alpha}''(s) = (-\cos s, -\sin s, 0)$$

At point $\vec{p}(s) = 0$ so that principal vector at $\vec{p}$ is

$$\vec{N}(0) = \vec{T}'(0) / \|\vec{T}'(0)\| = (-1, 0, 0)$$

Therefore

$$K_{n, \vec{U}}(\vec{v}) = \frac{\vec{U} \cdot \vec{N}}{R(\vec{v})} = \frac{-1}{1} = -1$$

References

- [1] Prentice .Hall,Inc ,Differential Geometry Of Curve And Surface ,New Jersey,1976 .
- [2] D.r:R.S .de Jong,Gaussian curvature And The Gauss-Bonnet Theorem,Bachelors thesis ,18 March 2013.
- [3] Gauss Curvature (Informal Treatment),6/9/2019.
- [4] Hindawi publishing corporation,Gaussian Curvature In Propagation Problems in Physics And Engineering , 2012.